NT-4c: Literature Extraction & Convention Fixing for FLRW Reduction of Nonlocal Field Equations

From Nonlinear SCT Field Equations to Modified Friedmann Equations and Cosmological Perturbation Framework

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Abstract

This document collects all formulas, conventions, and computational recipes from the literature needed to reduce the full nonlinear SCT field equations (NT-4b) to spatially flat FLRW symmetry and derive the modified Friedmann equations. The central simplification is that the Weyl tensor vanishes identically on conformally flat backgrounds ($C_{\mu\nu\rho\sigma} = 0$ on FLRW), so the entire C^2 sector governed by $F_1(\Box/\Lambda^2)$ drops out of the *background* equations. Only the R^2 sector, controlled by $\alpha_R(\xi) = 2(\xi - 1/6)^2$, contributes nonlocal corrections to the Friedmann equations. At conformal coupling $\xi = 1/6$, $\alpha_R = 0$ and the spectral action produces no background modifications beyond the Einstein–Hilbert term.

The document covers: (1) FLRW geometry and curvature identities; (2) the d'Alembertian on FLRW and its action on scalars; (3) reduction of the R^2 sector including the $\Theta^{(R)}_{\mu\nu}$ alpha-insertion terms; (4) the resulting modified Friedmann equations; (5) de Sitter self-consistency; (6) scalar, vector, and tensor perturbations on FLRW; (7) gravitational wave speed and the GW170817 constraint; (8) comparison with Deser–Woodard, Maggiore–Mancarella, Koshelev–Kumar–Starobinsky, and Biswas–Koshelev–Mazumdar nonlocal cosmological models. All formulas are extracted from the literature with source annotations; no original derivations are performed. Gaps requiring resolution in the derivation phase are noted explicitly.

Key references:

- NT-4b: Full nonlinear field equations (this project).
- NT-4a: Linearized field equations (this project).
- Stelle: Phys. Rev. D **16** (1977) 953; Gen. Rel. Grav. **9** (1978) 353.
- Barvinsky–Vilkovisky (BV): Phys. Rep. **119** (1985) 1; Nucl. Phys. B **282** (1987) 163; B **333** (1990) 471.
- Deser–Woodard (DW): Phys. Rev. Lett. **99** (2007) 111301 [1]; arXiv:1307.6639 [2]; arXiv:1902.08075 [3].
- Belgacem–Dirian–Foffa–Maggiore (BDFM): arXiv:1712.07066 [4].

- Dirian–Foffa–Khosravi–Kunz–Maggiore (DFKKM): arXiv:1403.6068 [5].
- Koshelev–Kumar–Starobinsky (KKS): arXiv:2305.18716 [7].
- Biswas–Mazumdar–Siegel (BMS): hep-th/0508194 [8].
- Biswas–Koivisto–Mazumdar (BKM): arXiv:1005.0590 [9].
- Briscese–Calcagni–Modesto (BCM): arXiv:1901.03267 [10].
- Calcagni–Modesto (CM): arXiv:2211.05606 [11].
- Nersisyan–Lima–Amendola (NLA): arXiv:1801.06683 [12].
- Belgacem–Dirian–Foffa–Maggiore: arXiv:1805.08731 [6].
- Nojiri–Odintsov (NO): arXiv:0810.1557 [13].

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1 Input: The NT-4b Field Equations

1.1 The Canonical Field Equations (NT-4b, Theorem 4.1)

The starting point for the FLRW reduction is the full nonlinear field equation derived in NT-4b:

$$\boxed{\frac{1}{\kappa^2}\,G_{\mu\nu} + \frac{1}{16\pi^2}\Big[2\,F_1(\Box/\Lambda^2)\,B_{\mu\nu} + \Theta^{(C)}_{\mu\nu}\Big] + \frac{1}{16\pi^2}\Big[F_2(\Box/\Lambda^2,\xi)\,H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\Big] = \frac{1}{2}\,T_{\mu\nu},} \tag{1}$$

where [NT-4b, Eq. (full-eom)]:

- $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$ is the Einstein tensor,
- $B_{\mu\nu} = \nabla^\rho\nabla^\sigma C_{\mu\rho\nu\sigma} + \frac{1}{2}R^{\rho\sigma}C_{\mu\rho\nu\sigma}$ is the Bach tensor (traceless, divergence-free),
- $H_{\mu\nu} = 2\nabla_\mu\nabla_\nu R - 2g_{\mu\nu}\Box R - \frac{1}{2}g_{\mu\nu}R^2 + 2R\,R_{\mu\nu}$ is the R^2-sector tensor with trace $g^{\mu\nu}H_{\mu\nu} = -6\Box R$,
- $\Theta^{(R)}_{\mu\nu}$ and $\Theta^{(C)}_{\mu\nu}$ are the BV alpha-insertion corrections,
- $\kappa^2 = 8\pi G$, Λ is the spectral cutoff.

1.2 Canonical Coefficients

From NT-1b Phase 3 [NT-1b, Phase 3 handoff]:

$$\alpha_C = 16\pi^2\,F_1(0) = \frac{13}{120}, \tag{2}$$

$$\alpha_R(\xi) = 16\pi^2\,F_2(0,\xi) = 2\Big(\xi - \frac{1}{6}\Big)^2, \tag{3}$$

$$c_2 = 2\alpha_C = \frac{13}{60}, \tag{4}$$

$$3c_1 + c_2 = 6\Big(\xi - \frac{1}{6}\Big)^2. \tag{5}$$

1.3 The $\Theta^{(R)}$ Alpha-Insertion (NT-4b, §4)

$$\Theta^{(R)}_{\mu\nu} = -\sum_{n=1}^{\infty} c_n^{(2)} \sum_{k=0}^{n-1}\Big[\nabla_\mu(\Box^k R)\,\nabla_\nu(\Box^{n-1-k}R) - \frac{1}{2}g_{\mu\nu}\,\nabla_\rho(\Box^k R)\,\nabla^\rho(\Box^{n-1-k}R)\Big], \tag{6}$$

where $F_2(z,\xi) = \sum_{n=0}^{\infty} c_n^{(2)}\,(z/\Lambda^2)^n$.

[Verified] NT-4b: 110/110 tests pass. Dual derivation (Methods A and B1) exact agreement. De Sitter, Bianchi, trace, local limit all verified.

2 FLRW Geometry

2.1 Metric and Symmetry

The spatially flat FLRW metric in cosmic time is [Weinberg (2008), §1.1; Mukhanov (2005), §5.1]:

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2\,\delta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j, \tag{7}$$

with scale factor $a(t)$, Hubble parameter $H \equiv \dot{a}/a$, and spatial curvature $k = 0$.

In conformal time τ defined by $\mathrm{d}\tau = \mathrm{d}t/a(t)$:

$$\mathrm{d}s^2 = a(\tau)^2\Big(-\mathrm{d}\tau^2 + \delta_{ij}\,\mathrm{d}x^i\,\mathrm{d}x^j\Big). \tag{8}$$

Convention 2.1 (Signature and time)**.** We use Lorentzian signature $(-,+,+,+)$ throughout this document, consistent with MR-1 (Wick rotation from Euclidean). Cosmic time derivatives are denoted by overdots, conformal time derivatives by primes.

2.2 Curvature Tensors on FLRW

The nonvanishing components of the Ricci tensor and Ricci scalar on the metric (7) are [Weinberg (2008), §1.5; Carroll (2004), §8.3]:

$$R_{00} = -3(\dot{H} + H^2) = -3\frac{\ddot{a}}{a}, \tag{9}$$

$$R_{ij} = \Big(\dot{H} + 3H^2\Big)\,a^2\,\delta_{ij} = \Big(a\ddot{a} + 2\dot{a}^2\Big)\,\delta_{ij}, \tag{10}$$

$$R = 6\Big(\dot{H} + 2H^2\Big) = 6\Big(\frac{\ddot{a}}{a} + H^2\Big). \tag{11}$$

The Riemann tensor on a spatially flat FLRW metric with $k = 0$ has components [Carroll (2004), §8.3; Weinberg (2008), §1.5]:

$$R^0{}_{i0j} = \frac{\ddot{a}}{a}\,\delta_{ij}, \qquad R^i{}_{jkl} = H^2\Big(\delta^i_k\,\delta_{jl} - \delta^i_l\,\delta_{jk}\Big), \tag{12}$$

where the spatial δ_{ij} is the coordinate Kronecker delta. Note: in Cartesian FLRW coordinates $\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j$, the component with all spatial indices lowered is $R_{0i0j} = g_{00}\,R^0{}_{i0j} = -\ddot{a}\,a\,\delta_{ij}$ (includes an extra factor of a^2 relative to the mixed component).

Proposition 2.2 (Weyl tensor on FLRW)**.** *The Weyl tensor vanishes identically on any FLRW spacetime:*

$$C_{\mu\nu\rho\sigma}\Big|_{\mathrm{FLRW}} = 0. \tag{13}$$

Proof. An FLRW spacetime is conformally flat (the metric is conformal to Minkowski space). The Weyl tensor is conformally invariant in $d = 4$ and vanishes on flat space, hence $C_{\mu\nu\rho\sigma}|_{\mathrm{FLRW}} = 0$. Alternatively, on a spacetime of the form $g_{\mu\nu} = a^2(\tau)\,\eta_{\mu\nu}$, the Weyl tensor vanishes because the Riemann tensor has the maximally symmetric structure. □

Critical Consequence: Weyl Sector Drops Out of Background

Since $C_{\mu\nu\rho\sigma} = 0$ on FLRW:

1. The Bach tensor vanishes: $B_{\mu\nu}\Big|_{\mathrm{FLRW}} = 0$.
2. The entire $F_1(\Box/\Lambda^2)$ sector does not contribute to the **background** Friedmann equations.
3. The alpha-insertion $\Theta^{(C)}_{\mu\nu}\Big|_{\mathrm{FLRW}} = 0$ since all terms involve $C_{\mu\nu\rho\sigma}$ or its derivatives.
4. Only the R^2 sector survives, governed by $\alpha_R(\xi) = 2(\xi - 1/6)^2$.
5. At conformal coupling $\xi = 1/6$: $\alpha_R = 0$, so all spectral corrections vanish and one recovers the standard Friedmann equations exactly.

2.3 The Ricci Tensor in Maximally Symmetric Form

On FLRW, the Ricci tensor takes the "perfect fluid" form:

$$R_{\mu\nu} = (\dot{H} + 3H^2)\, g_{\mu\nu} - 2\dot{H}\, u_\mu u_\nu, \tag{14}$$

where $u^\mu = (1, 0, 0, 0)$ is the comoving 4-velocity ($u_\mu = (-1, 0, 0, 0)$). One verifies: $R_{00} = -(\dot{H} + 3H^2) - 2\dot{H} \cdot 1 = -3(\dot{H} + H^2)$; $R_{ij} = (\dot{H} + 3H^2)\, a^2 \delta_{ij}$.

[Sign correction] LR audit correction: The original formula $R_{\mu\nu} = (R/4)\, g_{\mu\nu} + (R/4 + 3\dot{H})\, u_\mu u_\nu$ is incorrect on general FLRW. It holds only on de Sitter where $\dot{H} = 0$ and $R/4 = 3H^2 = \dot{H} + 3H^2$. On general FLRW, $R/4 = \frac{3}{2}(\dot{H} + 2H^2) \neq \dot{H} + 3H^2$.

On exact de Sitter ($\dot{H} = 0$, $R = 12H^2$):

$$R_{\mu\nu}\Big|_{\mathrm{dS}} = \frac{R}{4}\, g_{\mu\nu} = 3H^2\, g_{\mu\nu}. \tag{15}$$

3 The d'Alembertian on FLRW

3.1 Scalar d'Alembertian

For a scalar field $\phi(t)$ depending only on cosmic time [Weinberg (2008), §10.8; Mukhanov (2005), §5.2]:

$$\Box\phi = g^{\mu\nu}\nabla_\mu\nabla_\nu\phi = -\ddot{\phi} - 3H\dot{\phi}. \tag{16}$$

In conformal time:

$$\Box\phi = -\frac{1}{a^2}\Big(\phi'' + 2\mathcal{H}\phi'\Big), \tag{17}$$

where $\mathcal{H} = a'/a = aH$ is the conformal Hubble parameter.

3.2 $\Box R$ on FLRW

Since $R = R(t)$ on FLRW, applying (16):

$$\Box R = -\ddot{R} - 3H\dot{R}. \tag{18}$$

The explicit expression requires differentiating (11):

$$\dot{R} = 6(\ddot{H} + 4H\dot{H}), \tag{19}$$

$$\ddot{R} = 6\Big(\dddot{H} + 4\dot{H}^2 + 4H\ddot{H}\Big). \tag{20}$$

Remark 3.1 (De Sitter limit). On de Sitter ($H = \text{const}$): $\dot{R} = 0$, $\Box R = 0$, and $\Box^n R = 0$ for all $n \geq 1$.

3.3 Iterated $\Box^n R$

For the alpha-insertion terms (6), one needs $\Box^n R$ for $n = 1, 2, \ldots$. On FLRW each $\Box^n R$ is a function of t alone, computed iteratively:

$$\Box^{n+1} R = -\frac{\mathrm{d}^2}{\mathrm{d}t^2}(\Box^n R) - 3H\,\frac{\mathrm{d}}{\mathrm{d}t}(\Box^n R). \tag{21}$$

[GAP] The convergence of the infinite series in (6) on a general FLRW background must be established. The entireness of F_2 (NT-2) ensures $|c_n^{(2)}| \sim 1/n!$, which suggests convergence, but an explicit proof on FLRW is needed.

3.4 $F_2(\Box/\Lambda^2)$ Acting on R

The form factor acts on the Ricci scalar as:

$$F_2(\Box/\Lambda^2,\xi)\,R = \sum_{n=0}^{\infty} c_n^{(2)}\,\frac{\Box^n R}{\Lambda^{2n}}. \tag{22}$$

On de Sitter, this reduces to $F_2(\Box/\Lambda^2,\xi)\,R = c_0^{(2)}\,R_0 = F_2(0,\xi)\,R_0$, since $\Box^n R = 0$ for $n \geq 1$.

4 Reduction of the R^2 Sector to FLRW

4.1 The $H_{\mu\nu}$ Tensor on FLRW

The R^2-sector tensor (NT-4b, Eq. (H-tensor)) is:

$$H_{\mu\nu} = 2\nabla_\mu\nabla_\nu R - 2g_{\mu\nu}\Box R - \tfrac{1}{2}g_{\mu\nu}R^2 + 2R\,R_{\mu\nu}. \tag{23}$$

On FLRW with $R = R(t)$, each term evaluates as follows.

(i) $\nabla_\mu\nabla_\nu R$: Since $R = R(t)$, $\nabla_\mu R = \dot{R}\,\delta^0_\mu$ [standard: $\nabla_\mu R = \partial_\mu R$ for a scalar], and:

$$\nabla_0\nabla_0 R = \ddot{R}, \tag{24}$$
$$\nabla_i\nabla_j R = -H\dot{R}\,a^2\,\delta_{ij} = -H\dot{R}\,g_{ij}, \tag{25}$$
$$\nabla_0\nabla_i R = 0. \tag{26}$$

[Sign correction] LR audit correction: $\nabla_0 R = +\dot{R}$ (positive), not $-\dot{R}$. The raised-index gradient $\nabla^0 R = g^{00}\nabla_0 R = -\dot{R}$ has the minus sign; the lowered-index gradient does not. $\nabla_0\nabla_0 R = +\ddot{R}$ (via $\partial_0(\dot{R}) - \Gamma^\lambda_{00}\,\nabla_\lambda R = \ddot{R} - 0$). $\nabla_i\nabla_j R = -\Gamma^0_{ij}\,\nabla_0 R = -(a\dot{a}\,\delta_{ij})\dot{R} = -H\dot{R}\,a^2\delta_{ij}$ (negative, from the Christoffel connection).

(ii) Combining: The $(0,0)$ and (i,j) components of $H_{\mu\nu}$ on FLRW are (using $-2g_{00}\Box R = +2\Box R$, $-\frac{1}{2}g_{00}R^2 = +\frac{1}{2}R^2$):

$$\begin{aligned} H_{00} &= 2\ddot{R} + 2\Box R + \tfrac{1}{2}R^2 + 2R\cdot R_{00} \\ &= 2\ddot{R} + 2(-\ddot{R} - 3H\dot{R}) + \tfrac{1}{2}R^2 - 6R(\dot{H} + H^2) \\ &= -6H\dot{R} + \tfrac{1}{2}R^2 - 6R(\dot{H} + H^2), \end{aligned} \tag{27}$$

$$\begin{aligned} H_{ij} &= -2H\dot{R}\,a^2\delta_{ij} - 2\Box R\,a^2\delta_{ij} - \tfrac{1}{2}R^2\,a^2\delta_{ij} + 2R(\dot{H} + 3H^2)\,a^2\delta_{ij} \\ &= -2H\dot{R}\,a^2\delta_{ij} + 2(\ddot{R} + 3H\dot{R})\,a^2\delta_{ij} - \tfrac{1}{2}R^2\,a^2\delta_{ij} + 2R(\dot{H} + 3H^2)\,a^2\delta_{ij} \\ &= \left[2\ddot{R} + 4H\dot{R} + 2R(\dot{H} + 3H^2) - \tfrac{1}{2}R^2\right]a^2\delta_{ij}. \end{aligned} \tag{28}$$

[Sign correction] LR audit correction: The original computation had sign errors from $\nabla_0\nabla_0 R = -\ddot{R}$ and missing $g_{00} = -1$ sign flips. With $\nabla_0\nabla_0 R = +\ddot{R}$ and correct metric sign factors, H_{00} has $-6H\dot{R}$ and $+\frac{1}{2}R^2$ (signs flipped from original), and H_{ij} has coefficient $4H\dot{R}$ (not $8H\dot{R}$). Verified by the trace cross-check: $g^{\mu\nu}H_{\mu\nu} = -H_{00} + 3\cdot H_{ij}/(a^2) = -6\Box R$, which holds with these corrected signs but fails with the original ones.

Proposition 4.1 (De Sitter check)**.** *On de Sitter ($\dot{H} = 0$, $R = 12H^2 = \text{const}$): $\dot{R} = \ddot{R} = 0$, so*

$$H_{00}\Big|_{\text{dS}} = +\tfrac{1}{2}(12H^2)^2 - 6 \cdot 12H^2 \cdot H^2 = 72H^4 - 72H^4 = 0,$$
$$H_{ij}\Big|_{\text{dS}} = \left[2 \cdot 12H^2 \cdot 3H^2 - \tfrac{1}{2}(12H^2)^2\right] a^2\delta_{ij} = (72 - 72)\, H^4\, a^2\delta_{ij} = 0.$$

[Verified] De Sitter check: $H_{\mu\nu}|_{\text{dS}} = 0$ directly from the corrected formulas, consistent with NT-4b Check 5. The $+\frac{1}{2}R^2$ and $2R \cdot R_{00} = -6RH^2$ terms cancel exactly.

4.2 The $\Theta^{(R)}_{\mu\nu}$ Terms on FLRW

On FLRW, $R = R(t)$ implies $\nabla_\mu R = +\dot{R}\,\delta^0_\mu$, and more generally $\nabla_\mu(\Box^k R) = +\frac{\mathrm{d}}{\mathrm{d}t}(\Box^k R)\,\delta^0_\mu$. Therefore the alpha-insertion (6) gives (with $\nabla_\rho(\Box^k R)\,\nabla^\rho(\Box^{n-1-k}R) = \nabla_0 \cdot g^{00}\nabla_0 = -\dot{R}_k\,\dot{R}_{n-1-k}$):

$$\begin{aligned}\Theta^{(R)}_{00} &= -\sum_{n=1}^{\infty} c_n^{(2)} \sum_{k=0}^{n-1}\left[\dot{R}_k\,\dot{R}_{n-1-k} - \tfrac{1}{2}\dot{R}_k\,\dot{R}_{n-1-k}\right] \\ &= -\frac{1}{2}\sum_{n=1}^{\infty} c_n^{(2)} \sum_{k=0}^{n-1} \dot{R}_k\,\dot{R}_{n-1-k},\end{aligned} \tag{29}$$

$$\Theta^{(R)}_{ij} = -\frac{1}{2}\,a^2\delta_{ij} \sum_{n=1}^{\infty} c_n^{(2)} \sum_{k=0}^{n-1} \dot{R}_k\,\dot{R}_{n-1-k}, \tag{30}$$

where we abbreviated $R_k \equiv \Box^k R/\Lambda^{2k}$ and $\dot{R}_k \equiv \frac{\mathrm{d}}{\mathrm{d}t}(\Box^k R/\Lambda^{2k})$.

[Sign correction] LR audit correction: (1) In $\Theta^{(R)}_{00}$, the intermediate step uses "$-$" (not "$+$") for the second term in the bracket, because $-(1/2)\,g_{00}\,\nabla_\rho\nabla^\rho = -(1/2)(-1)(-\dot{R}_k\dot{R}_{n-1-k}) = -(1/2)\dot{R}_k\dot{R}_{n-1-k}$. The final result $-(1/2)\mathcal{S}$ is unchanged. (2) In $\Theta^{(R)}_{ij}$, the sign is $-1/2$ (not $+1/2$), because $\nabla_i(\Box^k R) = 0$ and the only contribution is $-(1/2)g_{ij}(-\dot{R}_k\dot{R}_{n-1-k}) = +(1/2)a^2\delta_{ij}\dot{R}_k\dot{R}_{n-1-k}$ inside the bracket, which is then multiplied by the overall $-\sum$ to give $-(1/2)a^2\delta_{ij}\mathcal{S}$.

Remark 4.2 (Structure of $\Theta^{(R)}$ on FLRW). On FLRW, $\Theta^{(R)}_{\mu\nu}$ takes a "perfect fluid" form:

$$\Theta^{(R)}_{\mu\nu} = (\rho_\Theta + p_\Theta)\,u_\mu u_\nu + p_\Theta\,g_{\mu\nu}, \tag{31}$$

with effective energy density and pressure:

$$\rho_\Theta = -\frac{1}{2}\mathcal{S}, \qquad p_\Theta = -\frac{1}{2}\mathcal{S}, \qquad \mathcal{S} \equiv \sum_{n=1}^{\infty} c_n^{(2)} \sum_{k=0}^{n-1} \dot{R}_k\,\dot{R}_{n-1-k}. \tag{32}$$

Note the stiff equation of state $p_\Theta = +\rho_\Theta$ ($w = +1$), characteristic of kinetic-energy terms involving time derivatives of R.

[GAP] The exact relation between the infinite series $\mathcal{S}$ and the nonlocal operator $F_2(\Box/\Lambda^2)$ acting on R needs to be made explicit. This connects to the integral representation of the alpha-insertion.

5 Modified Friedmann Equations

5.1 Standard Friedmann Equations

With a perfect fluid $T_{\mu\nu} = (\rho + p)\, u_\mu u_\nu + p\, g_{\mu\nu}$, the standard Friedmann equations are [Weinberg (2008), §1.5]:

$$H^2 = \frac{8\pi G}{3}\,\rho \quad \text{(first Friedmann)}, \tag{33}$$

$$\dot{H} + H^2 = -\frac{4\pi G}{3}(\rho + 3p) \quad \text{(acceleration / Raychaudhuri)}. \tag{34}$$

5.2 SCT-Modified Friedmann Equations

On FLRW, the full field equations (1) reduce to (using $B_{\mu\nu} = 0$, $\Theta^{(C)} = 0$):

$$\frac{1}{\kappa^2}\, G_{\mu\nu} + \frac{1}{16\pi^2}\Big[F_2(\Box/\Lambda^2, \xi)\, H_{\mu\nu} + \Theta^{(R)}_{\mu\nu}\Big] = \frac{1}{2}\, T_{\mu\nu}. \tag{35}$$

Taking the $(0,0)$ component and the trace, we obtain the modified Friedmann equations.

Theorem 5.1 (Modified first Friedmann equation)**.** *The* $(0,0)$ *component of* (35) *gives:*

$$3H^2 + \frac{\kappa^2}{16\pi^2}\Big[F_2(\Box/\Lambda^2, \xi)\, H_{00} + \Theta^{(R)}_{00}\Big] = \kappa^2\, \rho, \tag{36}$$

where H_{00} *is given by* (27) *and* $\Theta^{(R)}_{00}$ *by* (29)*.*

Theorem 5.2 (Modified trace equation)**.** *Taking the trace of* (35)*:*

$$-\frac{R}{\kappa^2} + \frac{1}{16\pi^2}\Big[-6\, F_2(\Box/\Lambda^2, \xi)\, \Box R + g^{\mu\nu}\Theta^{(R)}_{\mu\nu}\Big] = \frac{1}{2}\,(-\rho + 3p), \tag{37}$$

using $g^{\mu\nu}G_{\mu\nu} = -R$*,* $g^{\mu\nu}H_{\mu\nu} = -6\Box R$ *(NT-4b Prop. 3.1), and* $g^{\mu\nu}\Theta^{(R)}_{\mu\nu} = -\rho_\Theta + 3p_\Theta = 2\rho_\Theta = -\mathcal{S}$ *from* (32) *(using* $p_\Theta = \rho_\Theta$*,* $w = +1$*).*

5.3 Local Limit: $R + R^2$ Gravity (Starobinsky)

In the local limit $F_2 \to F_2(0, \xi) = \alpha_R(\xi)/(16\pi^2)$, $\Theta^{(R)} = 0$, the trace equation reduces to:

$$R + \frac{6\alpha_R(\xi)}{16\pi^2\kappa^{-2}}\,\Box R = -\kappa^2\,(\rho - 3p)/2. \tag{38}$$

Comparing with Starobinsky's $R + R^2/(6M^2)$ model [Starobinsky (1980); Mukhanov & Chibisov (1981)]: the correspondence is $M^2 = 16\pi^2/(6\alpha_R(\xi)\kappa^2) = 1/(6\alpha_R(\xi)\cdot 8\pi G)$.

5.4 De Sitter Self-Consistency

On de Sitter ($H = \text{const}$, $R = 12H^2$):

- $\dot{R} = 0$, $\Box R = 0$, $\Box^n R = 0$ for all $n \geq 1$.
- $H_{\mu\nu}|_{\text{dS}} = 0$ (NT-4b, Prop. 4.2).

- $\Theta^{(R)}_{\mu\nu}|_{\rm dS} = 0$ (all terms involve ∇R).

Therefore the modified Friedmann equation (36) reduces to $3H^2 = \kappa^2\rho$, i.e., standard Einstein gravity. De Sitter is an exact vacuum solution for $\rho = p = 0$ with $H^2 = \Lambda_{\rm eff}/3$.

[Verified] This is consistent with NT-4b Check 5 (de Sitter exact solution). The FLRW reduction inherits this check automatically.

6 Comparison with Nonlocal Cosmological Models

6.1 Deser–Woodard Model

The DW model [1, 2, 3] modifies gravity via:

$$S_{\rm DW} = \frac{1}{16\pi G}\int {\rm d}^4x\,\sqrt{-g}\left[R + R\,f\left(\frac{R}{\Box}\right)\right], \tag{39}$$

where f is a dimensionless function. The model introduces no additional mass scale (both R and $\Box$ have dimension $[\text{length}]^{-2}$).

Differences from SCT:

1. DW uses $\Box^{-1}$ (retarded Green's function), which is infrared-motivated. SCT uses entire form factors $F_i(\Box/\Lambda^2)$, which are UV-motivated.
2. DW modifies gravity in the IR (dark energy). SCT modifies it at the spectral cutoff scale Λ.
3. DW has $c_T = c$ (GWs travel at the speed of light) [12]. SCT needs verification (see §7).

Key result [Deser–Woodard (2013), §2]: On FLRW, the DW model introduces an auxiliary field $X \equiv \Box^{-1}R$ satisfying $\Box X = R$, which localizes the dynamics. The modified Friedmann equations involve $X(t)$ and its derivatives, producing an effective dark energy component.

6.2 Maggiore–Mancarella RR Model

The RR model [4, 5] adds $m^2\,R\,\Box^{-2}\,R$ to the Einstein–Hilbert action:

$$S_{\rm RR} = \frac{1}{16\pi G}\int {\rm d}^4x\,\sqrt{-g}\left[R - \frac{m^2}{6}\,R\,\frac{1}{\Box^2}\,R\right]. \tag{40}$$

Key results [Belgacem et al. (2018), §3]:

- The model introduces two auxiliary fields $U = -\Box^{-1}R$ and $S = -\Box^{-1}U$ to localize the dynamics.
- On FLRW, the modified Friedmann equation becomes:

$$H^2 = \frac{8\pi G}{3}\rho + \frac{m^2}{6}\Big[\text{terms involving } U, S, \dot{U}, \dot{S}\Big]. \tag{41}$$

- The dark energy equation of state is phantom: $w_{\rm DE}(0) \approx -1.14$.
- GWs propagate at the speed of light: $c_T = c$.
- The model reduces H_0 tension to 2.0σ.

Differences from SCT: The RR model uses $\Box^{-2}$ (IR nonlocality), while SCT uses entire functions (UV nonlocality). The RR model has one free parameter m replacing Ω_Λ; SCT has Λ (spectral cutoff) and ξ (Higgs non-minimal coupling).

6.3 Koshelev–Kumar–Starobinsky Nonlocal Inflation

The KKS approach [7] considers the action:

$$S_{\text{KKS}} = \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{-g}\,\Big[M_p^2\,R + R\,\mathcal{F}_R(\Box_s)\,R + W_{\mu\nu\rho\sigma}\,\mathcal{F}_C(\Box_s)\,W^{\mu\nu\rho\sigma}\Big], \tag{42}$$

where $\Box_s = \Box/\mathcal{M}_s^2$ and $\mathcal{F}_R$, $\mathcal{F}_C$ are analytic infinite-derivative (AID) operators.

Key results [KKS (2023), §2–3]:

- On FLRW, the Weyl sector drops out (same as SCT).
- The trace equation reduces to $\Box R = M^2 R$, the Starobinsky scalaron equation, under the condition $\mathcal{F}_R(M^2/\mathcal{M}_s^2) = M_p^2/(6M^2)$.
- Ghost-free form factors require $\mathcal{O}_0 = (1 - \Box/M^2)\,e^{H_0(\Box_s)}$ and $\mathcal{O}_2 = e^{H_2(\Box_s)}$ where H_0, H_2 are entire functions.
- The form factors are *background-dependent* on curved spacetimes, necessitating cubic and higher curvature terms for a background-independent formulation.
- On FLRW, the scalaron equation of motion provides R^2-like inflation.

Relevance to SCT: The SCT form factors F_1, F_2 are derived from the spectral action, not postulated. The KKS ghost-freedom conditions correspond to the SCT requirement that F_1, F_2 are entire (NT-2). The scalaron mass in KKS corresponds to the SCT effective mass $m_0(\xi) = \Lambda/\sqrt{6(\xi - 1/6)^2}$.

6.4 Biswas–Koshelev–Mazumdar Bouncing Cosmology

The BKM approach [8, 9] obtains bouncing cosmological solutions from ghost-free nonlocal gravity:

$$S_{\text{BKM}} = \frac{M_p^2}{2}\int \mathrm{d}^4x\,\sqrt{-g}\,\Big[R + R\,\mathcal{F}(\Box)\,R\Big]. \tag{43}$$

Key results [BKM (2010), §3]:

- Stable inflationary attractor solutions exist with positive cosmological constant.
- Cyclic universe solutions emerge with negative cosmological constant.
- Super-Hubble perturbations are tracked through the bounce.
- Bouncing solutions are free from perturbative instability.

Relevance to SCT: The BKM action is structurally identical to the R^2 sector of the SCT spectral action, with $\mathcal{F}(\Box) \leftrightarrow F_2(\Box/\Lambda^2, \xi)/(16\pi^2)$. The SCT form factor F_2 is derived from first principles (spectral action + heat kernel), while BKM postulates $\mathcal{F}$.

7 Gravitational Wave Speed

7.1 The GW170817 Constraint

The joint detection of GW170817 and GRB 170817A established [LIGO–Virgo Collaboration (2017)]:

$$\left|c_T/c - 1\right| \leq 1 \times 10^{-15}, \tag{44}$$

where c_T is the speed of gravitational waves. This constraint applies at $z \approx 0$ (the source was at 40 Mpc).

7.2 Tensor Perturbations on FLRW

Consider tensor perturbations h_{ij}^{TT} on the FLRW background [Mukhanov (2005), §7; Maggiore (2008), §19]:

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\left(\delta_{ij} + h_{ij}^{\mathrm{TT}}\right)\mathrm{d}x^i\,\mathrm{d}x^j, \tag{45}$$

where h_{ij}^{TT} is transverse and traceless: $\partial_i h_{\mathrm{TT}}^{ij} = 0$, $h^i{}_i = 0$.

In GR, the tensor perturbation equation is:

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{\nabla^2}{a^2}\,h_{ij} = 0. \tag{46}$$

7.3 GW Propagation in SCT on FLRW

Critical: The Weyl Sector Contributes to Perturbations

Although $C_{\mu\nu\rho\sigma} = 0$ on the FLRW *background*, the perturbation $\delta C_{\mu\nu\rho\sigma} \neq 0$ for tensor modes. Therefore the F_1 (Weyl) sector *does* contribute to the tensor perturbation equation, even though it does not appear in the background Friedmann equations.

This is a standard feature of higher-derivative gravity theories: the linearized equation of motion for perturbations around a curved background can involve terms from sectors that vanish on the background itself.

From the NT-4b field equations linearized on FLRW, the tensor perturbation equation takes the schematic form:

$$\left[1 + \mathcal{G}_T(\Box/\Lambda^2)\right]\left(\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{\nabla^2}{a^2}\,h_{ij}\right) = 0, \tag{47}$$

where $\mathcal{G}_T$ encodes the nonlocal modification from the Weyl sector.

[GAP] The exact form of $\mathcal{G}_T$ on a general FLRW background needs to be derived. On flat background, it reduces to the NT-4a result $\Pi_{\mathrm{TT}}(z) = 1 + (13/60)\,z\,\hat{F}_1(z)$.

The GW speed: For a mode $h_{ij} \propto e^{i(\omega t - \mathbf{k}\cdot\mathbf{x})}$, the dispersion relation from (47) gives:

$$\omega^2 = \frac{k^2}{a^2} \cdot c_T^2(k,t), \tag{48}$$

where c_T depends on k in general (nonlocal dispersion).

Low-k limit: In the limit $k \to 0$ (long wavelengths, relevant for GW170817):

$$c_T^2(k \to 0) = 1, \tag{49}$$

because the form factors reduce to their $z = 0$ values and the effective action becomes the local Einstein–Hilbert action (plus local curvature-squared terms that do not modify the GW speed in $d = 4$).

Proposition 7.1 (GW speed in SCT at low momenta)**.** *At momenta $k \ll \Lambda$ (equivalently $z = k^2/\Lambda^2 \ll 1$), the SCT prediction for the GW speed on FLRW is $c_T = c$, consistent with the GW170817 bound (44).*

This follows from the fact that:

1. In the local limit ($z \to 0$), the Weyl sector gives $2F_1(0)\, B_{\mu\nu} = (13/60) \cdot B_{\mu\nu}/(16\pi^2)$.
2. The local C^2 term does not modify the GW speed in 4D [Stelle (1977); Nersisyan et al. (2018), §3].
3. The R^2 term does not affect tensor perturbations at leading order (it couples to the scalar sector only).

Comparison: Both the DW model [12] and the Maggiore RR model [4] have $c_T = c$ exactly. The Horndeski models with $G_4(X) \neq 0$ or $G_5 \neq 0$ are excluded [12]. The SCT prediction $c_T(k \to 0) = c$ places it in the same class as DW and RR.

[GAP] The high-k behavior of $c_T(k)$ (UV dispersion) is determined by $F_1(z)$ and may show deviations from c at $k \sim \Lambda$. This is expected from the MR-3 causality analysis (micro-causality violated at $\ell \sim 1/\Lambda$).

8 Cosmological Perturbation Theory on FLRW

8.1 SVT Decomposition

The perturbed FLRW metric in the Newtonian (longitudinal) gauge is [Mukhanov (2005), §7; Bardeen (1980)]:

$$\mathrm{d}s^2 = -(1 + 2\Phi)\, \mathrm{d}t^2 + a^2(t)\Big[(1 - 2\Psi)\, \delta_{ij} + h_{ij}^{\mathrm{TT}}\Big]\, \mathrm{d}x^i\, \mathrm{d}x^j, \tag{50}$$

where Φ and Ψ are the Bardeen scalar potentials and h_{ij}^{TT} is the transverse-traceless tensor perturbation. Vector perturbations decay in an expanding universe and are omitted.

8.2 Scalar Perturbations and Gravitational Slip

In GR, $\Phi = \Psi$ in the absence of anisotropic stress. In modified gravity, the gravitational slip parameter [Amendola et al. (2008)]:

$$\eta \equiv -\frac{\Phi}{\Psi} \tag{51}$$

can differ from unity. For the DW model, $\eta \approx -0.05 + 1.79\, a^2 - 0.54\, a^3$ [12].

In SCT on FLRW, the R^2 sector generates anisotropic stress via the $H_{\mu\nu}$ tensor, producing $\Phi \neq \Psi$. The Weyl sector F_1 also contributes to scalar perturbations through $\delta C_{\mu\nu\rho\sigma}$.

[GAP] The explicit expressions for $\eta(k, z)$ in SCT need to be derived in the NT-4c derivation phase. This is important for INF-1 (n_s, r) and MT-2 (H_0, S_8 tensions).

8.3 Inflationary Observables (Preview for INF-1)

For R^2-like inflation driven by the F_2 sector, the key observables are [KKS (2023), §4]:

$$n_s = 1 - \frac{2}{N} \quad \text{(spectral index, at leading order)}, \tag{52}$$

$$r = \frac{12}{N^2} \quad \text{(tensor-to-scalar ratio, at leading order)}, \tag{53}$$

where N is the number of e-folds. For $N = 55$: $n_s \approx 0.964$, $r \approx 0.004$, consistent with Planck 2018 constraints $n_s = 0.9649 \pm 0.0042$ and $r < 0.064$ (95% CL).

In the SCT framework, nonlocal corrections from $F_2(\Box/\Lambda^2)$ modify both n_s and r at sub-leading order. These corrections depend on the ratio $M^2/\mathcal{M}_s^2$ (scalaron mass vs. nonlocality scale).

[GAP] The computation of n_s and r in SCT, including nonlocal corrections, is deferred to INF-1.

9 Notation Table

Symbol	Definition / Convention
$g_{\mu\nu}$	Lorentzian metric, signature $(-,+,+,+)$
$a(t)$	Scale factor
$H = \dot{a}/a$	Hubble parameter
$\mathcal{H} = a'/a$	Conformal Hubble parameter
τ	Conformal time: $\mathrm{d}\tau = \mathrm{d}t/a$
R	Ricci scalar: $6(\dot{H} + 2H^2)$ on FLRW
$\Box$	d'Alembertian: $-\ddot{} - 3H\dot{}$ for scalars on FLRW
Λ	Spectral cutoff scale
ξ	Higgs non-minimal coupling
κ^2	$8\pi G$
α_C	13/120 (Weyl coefficient)
$\alpha_R(\xi)$	$2(\xi - 1/6)^2$ (Ricci-scalar coefficient)
$F_1(z)$	Weyl form factor, $z = \Box/\Lambda^2$
$F_2(z,\xi)$	R^2 form factor
$B_{\mu\nu}$	Bach tensor (from $\delta C^2/\delta g^{\mu\nu}$)
$H_{\mu\nu}$	R^2-sector tensor
$\Theta^{(R)}_{\mu\nu}$	BV alpha-insertion, R^2 sector
$\Theta^{(C)}_{\mu\nu}$	BV alpha-insertion, Weyl sector
Φ, Ψ	Bardeen scalar potentials
h^{TT}_{ij}	Tensor perturbation
c_T	Gravitational wave speed
n_s	Scalar spectral index
r	Tensor-to-scalar ratio
N	Number of e-folds

10 Summary of Gaps for the Derivation Phase

#	Gap	Description	Priority
G1	$H_{\mu\nu}$ on FLRW	Verify the explicit $(0,0)$ and (i,j) components of $H_{\mu\nu}$ on FLRW, including the action of $F_2(\Box/\Lambda^2)$.	HIGH
G2	$\Theta^{(R)}$ convergence	Prove convergence of the alpha-insertion series on FLRW. Establish connection between the series $\mathcal{S}$ and the integral representation.	HIGH
G3	Tensor perturbation equation	Derive the exact form of $\mathcal{G}_T(\Box/\Lambda^2)$ on FLRW from the linearization of the Weyl sector around the FLRW background. Extract $c_T(k)$.	HIGH
G4	Scalar perturbation equations	Derive the coupled equations for Φ, Ψ from the linearization of both the R^2 and Weyl sectors on FLRW. Extract $\eta(k,z)$.	MEDIUM
G5	Eigenvalue ansatz	Test the KKS ansatz $\Box R = M^2 R$ in the SCT context. Determine whether the SCT form factor F_2 admits this simplification on the FLRW background.	MEDIUM
G6	Conformal coupling limit	At $\xi = 1/6$, $\alpha_R = 0$ and the R^2 sector vanishes. Verify that the tensor perturbation equation still receives contributions from the Weyl sector F_1 (which has $\alpha_C = 13/120 \neq 0$).	HIGH
G7	Energy conditions	Check whether the effective energy density and pressure from the nonlocal terms satisfy standard energy conditions (null, weak, strong).	LOW

11 Benchmarks for the Derivation Phase

The following benchmarks must be satisfied by any correct derivation:

B1 De Sitter: The modified Friedmann equations must reduce to $3H^2 = \kappa^2\rho$ on de Sitter, consistent with NT-4b Check 5.

B2 Local limit: Setting $F_2 \to F_2(0,\xi)$, $\Theta^{(R)} \to 0$, the equations must reproduce the local $R + \beta R^2$ Friedmann equations with $\beta = \alpha_R(\xi)/(16\pi^2)$.

B3 Conformal coupling: At $\xi = 1/6$, $\alpha_R = 0$, and the *background* Friedmann equations must reduce to standard GR.

B4 GW speed: $c_T(k \to 0) = c$ at low momenta, consistent with GW170817.

B5 Linearized recovery: On flat background ($a = 1$, $H = 0$), the perturbation equations must reproduce NT-4a Π_{TT} and Π_{s}.

B6 Energy conservation: The modified equations must be consistent with $\nabla^\mu T_{\mu\nu} = 0$ (Bianchi identity).

B7 Trace consistency: The trace of the modified Friedmann equations must give the Raychaudhuri equation with the same nonlocal corrections.

B8 KKS scalaron equation: Under the ansatz $\Box R = M^2 R$, the SCT trace equation must reduce to the Starobinsky scalaron equation.

References

[1] S. Deser and R. P. Woodard, "Nonlocal cosmology," *Phys. Rev. Lett.* **99** (2007) 111301, arXiv:0706.2151.

[2] S. Deser and R. P. Woodard, "Observational viability and stability of nonlocal cosmology," *JCAP* **1311** (2013) 036, arXiv:1307.6639.

[3] S. Deser and R. P. Woodard, "Nonlocal cosmology II — Cosmic acceleration without fine tuning or dark energy," *JCAP* **1906** (2019) 034, arXiv:1902.08075.

[4] E. Belgacem, Y. Dirian, S. Foffa, and M. Maggiore, "Nonlocal gravity. Conceptual aspects and cosmological predictions," *JCAP* **1803** (2018) 002, arXiv:1712.07066.

[5] Y. Dirian, S. Foffa, N. Khosravi, M. Kunz, and M. Maggiore, "Cosmological perturbations and structure formation in nonlocal infrared modifications of general relativity," *JCAP* **1406** (2014) 033, arXiv:1403.6068.

[6] E. Belgacem, Y. Dirian, S. Foffa, and M. Maggiore, "Modified gravitational-wave propagation and standard sirens," *Phys. Rev. D* **98** (2018) 023510, arXiv:1805.08731.

[7] A. S. Koshelev, K. Sravan Kumar, and A. A. Starobinsky, "Cosmology in nonlocal gravity," in *Handbook of Quantum Gravity*, Springer, 2023, arXiv:2305.18716.

[8] T. Biswas, A. Mazumdar, and W. Siegel, "Bouncing universes in string-inspired gravity," *JCAP* **0603** (2006) 009, hep-th/0508194.

[9] T. Biswas, T. Koivisto, and A. Mazumdar, "Towards a resolution of the cosmological singularity in non-local higher derivative theories of gravity," *JCAP* **1011** (2010) 008, arXiv:1005.0590.

[10] F. Briscese, G. Calcagni, and L. Modesto, "Nonlinear stability in nonlocal gravity," *Phys. Rev. D* **99** (2019) 084041, arXiv:1901.03267.

[11] A. Bas i Beneito, G. Calcagni, and L. Rachwał, "Classical and quantum nonlocal gravity," in *Handbook of Quantum Gravity*, Springer, 2022, arXiv:2211.05606.

[12] H. Nersisyan, N. A. Lima, and L. Amendola, "Gravitational wave speed: Implications for models without a mass scale," *Phys. Lett. B* **783** (2018) 188–193, arXiv:1801.06683.

[13] S. Nojiri and S. D. Odintsov, "Modified gravity as realistic candidate for dark energy, inflation and dark matter," arXiv:0810.1557.

[14] K. S. Stelle, "Renormalization of higher-derivative quantum gravity," *Phys. Rev. D* **16** (1977) 953–969.

[15] K. S. Stelle, "Classical gravity with higher derivatives," *Gen. Rel. Grav.* **9** (1978) 353–371.

[16] A. O. Barvinsky and G. A. Vilkovisky, "The generalized Schwinger-DeWitt technique in gauge theories and quantum gravity," *Phys. Rep.* **119** (1985) 1–74; *Nucl. Phys. B* **282** (1987) 163–188.

[17] A. O. Barvinsky and G. A. Vilkovisky, "Covariant perturbation theory (II): second order in the curvature," *Nucl. Phys. B* **333** (1990) 471–511.

[18] A. A. Starobinsky, "A new type of isotropic cosmological models without singularity," *Phys. Lett. B* **91** (1980) 99–102.

[19] A. Codello, R. Percacci, and C. Rahmede, "Investigating the ultraviolet properties of gravity with a Wilsonian renormalization group equation," *Ann. Phys.* **324** (2009) 414–469, arXiv:0805.2909.

[20] A. Codello and O. Zanusso, "On the non-local heat kernel expansion," *J. Math. Phys.* **54** (2013) 013513, arXiv:1203.2034.

[21] B. P. Abbott *et al.* (LIGO Scientific Collaboration and Virgo Collaboration), "Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A," *Astrophys. J. Lett.* **848** (2017) L13.

[22] V. F. Mukhanov, *Physical Foundations of Cosmology*, Cambridge University Press, 2005.

[23] S. Weinberg, *Cosmology*, Oxford University Press, 2008.

[24] S. M. Carroll, *Spacetime and Geometry: An Introduction to General Relativity*, Addison-Wesley, 2004.

[25] J. M. Bardeen, "Gauge-invariant cosmological perturbations," *Phys. Rev. D* **22** (1980) 1882–1905.